CHEMISTRY 13 — 20 CARD FROM (A01–A10 & B01–B10)

Word Document Templates based on Structure→Name and Name→Structure Specifications

PART 1: STRUCTURE → NAME CARDS (A01 to A10)

CARD A01 — STRUCTURE → NAME

Level: Introductory | Molecular formula (counted):: C6H14

1. NUMBERED PARENT

Straight chain of 6 carbons (hexane). Numbered left-to-right (1 to 6).

2. NAMING TRAIL

Parent: hexane
Numbering: equivalent both ways
Substituents: none

3. MY NAME (WRITTEN FIRST)

hexane

4. THE ALKANE CHECK

6 carbons -> C6H2(6)+2 = C6H14
✓ Matches structure and formula.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [ ] formula counted

CARD A02 — STRUCTURE → NAME

Level: Easy | Molecular formula (counted):: C6H14

1. NUMBERED PARENT

5-carbon longest chain with a methyl group at C2 (left-to-right).

2. NAMING TRAIL

Parent: pentane
Numbering: left-to-right gives locant 2
Substituents: 2-methyl

3. MY NAME (WRITTEN FIRST)

2-methylpentane

4. THE ALKANE CHECK

5 (chain) + 1 (methyl) = 6 carbons -> C6H14
✓ Lowest locant chosen.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A03 — STRUCTURE → NAME

Level: Easy | Molecular formula (counted):: C7H16

1. NUMBERED PARENT

5-carbon main chain with two methyl groups at C2 and C3.

2. NAMING TRAIL

Parent: pentane
Numbering: left-to-right (vs 3,4-dimethyl from right)
Substituents: 2,3-dimethyl

3. MY NAME (WRITTEN FIRST)

2,3-dimethylpentane

4. THE ALKANE CHECK

5 + 2 = 7 carbons -> C7H16
✓ Lowest locant set selected.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A04 — STRUCTURE → NAME

Level: Medium | Molecular formula (counted):: C8H18

1. NUMBERED PARENT

5-carbon main chain with a methyl at C2 and an ethyl at C3.

2. NAMING TRAIL

Parent: pentane
Numbering: 3-ethyl-2-methyl
Alphabetical order: ethyl before methyl

3. MY NAME (WRITTEN FIRST)

3-ethyl-2-methylpentane

4. THE ALKANE CHECK

5-carbon chain + ethyl (2) + methyl (1) = 8 carbons -> C8H18 ✓

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A05 — STRUCTURE → NAME

Level: Medium | Molecular formula (counted):: C8H18

1. NUMBERED PARENT

6-carbon main chain (hexane) with two methyl groups at C2 and C4.

2. NAMING TRAIL

Parent: hexane •
Numbering: left-to-right gives locants 2,4 (vs. right-to-left giving 3,5).
Substituents: 2,4-dimethyl

3. MY NAME (WRITTEN FIRST)

2,4-dimethylhexane

4. THE ALKANE CHECK

6 + 2 = 8 carbons -> C8H18
✓ Lowest locant rule applied.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A06 — STRUCTURE → NAME

Level: Medium | Molecular formula (counted):: C9H18

1. NUMBERED PARENT

6-carbon ring (cyclohexane) with an isopropyl substituent at C1.

2. NAMING TRAIL

Parent: cyclohexane
Numbering: ring carbon with substituent is C1
Substituents: isopropyl

3. MY NAME (WRITTEN FIRST)

isopropylcyclohexane

4. THE ALKANE CHECK

Ring (6) + substituent (3) = 9 carbons -> C9H18 ✓
Ring naming verified.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A07 — STRUCTURE → NAME

Level: Hard | Molecular formula (counted):: C6H11Br

1. NUMBERED PARENT

6-carbon ring (cyclohexane) with a bromo group and a methyl group.

2. NAMING TRAIL

Parent: cyclohexane
Numbering: give lowest locants2 giving priority alphabetically (Bromo is lower locant 1, methyl is 2).
Substituents: 1-bromo-2-methyl

3. MY NAME (WRITTEN FIRST)

1-bromo-2-methylcyclohexane

4. THE ALKANE/HALIDE CHECK

C6H11Br ✓ Alphabetical order ('b' before 'm') and lowest locants satisfied.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A08 — STRUCTURE → NAME

Level: Hard | Molecular formula (counted):: C5H10Cl2

1. NUMBERED PARENT

5-carbon chain (pentane) with two chloro substituents at C2 and C4.

2. NAMING TRAIL

Parent: pentane
Numbering: left-to-right gives 2,4-dichloropentane (symmetric numbering gives same).
Substituents: 2,4-dichloro

3. MY NAME (WRITTEN FIRST)

2,4-dichloropentane

4. THE ALKANE CHECK

C5H10Cl2 ✓ Lowest locant set (2,4) correctly assigned.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A09 — STRUCTURE → NAME

Level: Hard | Molecular formula (counted):: C7H15Br

1. NUMBERED PARENT

6-carbon chain (hexane) with bromo and methyl groups at C3.

2. NAMING TRAIL

Parent: hexane
Numbering: left-to-right gives locant 3 for both
Substituents: 3-bromo-3-methylhexane

3. MY NAME (WRITTEN FIRST)

3-bromo-3-methylhexane

4. THE ALKANE CHECK

C7H15Br ✓ Alphabetical order ('bromo' before 'methyl') respected.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

CARD A10 — STRUCTURE → NAME

Level: Expert | Molecular formula (counted):: C7H14

1. NUMBERED PARENT

5-carbon ring (cyclopentane) with two methyl groups at C1 and C3.

2. NAMING TRAIL

Parent: cyclopentane
Numbering: 1,3-dimethyl (lowest possible locant set for ring)
Substituents: 1,3-dimethyl

3. MY NAME (WRITTEN FIRST)	4. THE ALKANE CHECK
1,3-dimethylcyclopentane	Ring (5) + 2 methyls (2) = 7 carbons -> C7H14 ✓ Ring numbering verified.

VERIFICATION STRIP

[✓] longest chain/ring [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

PART 2: NAME → STRUCTURE CARDS (B01 to B10)

CARD B01 — NAME → STRUCTURE

Level: Introductory | Given name:: propane

1. NAME DECODED	2. BOND-LINE STRUCTURE (THE ALKANE)
Parent: propane (3 carbons) Substituents: none Expected formula: C3H8	V-shape single line (3 connected vertices: CH3-CH2-CH3)
3. FORMULA CHECK	**4. BACK-CHECK**
Counted: C3H8 vs CnHz+2 (C3H8) -> Matches.	Returned name: propane ✓ / Correct. Parent chain length is 3.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B02 — NAME → STRUCTURE

Level: Easy | Given name:: 2-methylbutane

1. NAME DECODED	2. BOND-LINE STRUCTURE (THE ALKANE)
Parent: butane (4 carbons) Substituents: 2-methyl Expected formula: C5H12	4-carbon zig-zag backbone with a methyl branch pointing up/down from carbon 2.
3. FORMULA CHECK	**4. BACK-CHECK**
Counted: C5H12 vs CnHz+2 (C5H12) -> Matches.	Returned name: 2-methylbutane ✓ / Correct. Longest chain is 4, substituent at C2.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B03 — NAME → STRUCTURE

Level: Easy | Given name:: 2,2-dimethylpropane

1. NAME DECODED

Parent: propane (3 carbons)
Substituents: two methyls at C2
Expected formula: C5H12

2. BOND-LINE STRUCTURE (THE ALKANE)

Central carbon (C2) with 4 single bonds (cross layout representing neopentane).

3. FORMULA CHECK

Counted: C5H12 vs CnHz+2 (C5H12) -> Matches.

4. BACK-CHECK

Returned name: 2,2-dimethylpropane ✓ / Correct.
Longest chain is 3 with geminal methyls at C2.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B04 — NAME → STRUCTURE

Level: Medium | Given name:: 3-ethylhexane

1. NAME DECODED

Parent: hexane (6 carbons)
Substituents: 3-ethyl
Expected formula: C8H18

2. BOND-LINE STRUCTURE (THE ALKANE)

6-carbon zig-zag chain with a 2-carbon ethyl branch branching off carbon 3.

3. FORMULA CHECK

Counted: C8H18 vs CnHz+2 (C8H18) -> Matches.

4. BACK-CHECK

Returned name: 3-ethylhexane ✓ / Correct. Longest continuous chain is 6 carbons.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B05 — NAME → STRUCTURE

Level: Medium | Given name:: 4-ethyl-3-methylheptane

1. NAME DECODED

Parent: heptane (7 carbons)
Substituents: 4-ethyl, 3-methyl
Expected formula: C10H22

2. BOND-LINE STRUCTURE (THE ALKANE)

7-carbon chain with methyl at C3 and ethyl at C4.

3. FORMULA CHECK

Counted: C10H22 vs CnHz+2 (C10H22) -> Matches.

4. BACK-CHECK

Returned name: 4-ethyl-3-methylheptane ✓ / Correct. Alphabetical order and locants verified.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B06 — NAME → STRUCTURE

Level: Medium | Given name:: cyclopentane

1. NAME DECODED

Parent: cyclopentane (5-carbon ring)
Substituents: none
Expected formula: C5H10

2. BOND-LINE STRUCTURE (THE ALKANE)

Pentagon ring shape.

3. FORMULA CHECK

Counted: C5H10 vs CnH2n (C5H10) -> Matches ring formula.

4. BACK-CHECK

Returned name: cyclopentane ✓ / Correct. 5 carbons in a closed ring.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B07 — NAME → STRUCTURE

Level: Hard | Given name:: 1,1-dimethylcyclohexane

1. NAME DECODED

Parent: cyclohexane (6-carbon ring)
Substituents: two methyls at C1
Expected formula: C8H16

2. BOND-LINE STRUCTURE (THE ALKANE)

Hexagon ring with two single-bond lines pointing out from the same ring vertex (C1).

3. FORMULA CHECK

Counted: C8H16 vs CnH2n (C8H16) -> Matches.

4. BACK-CHECK

Returned name: 1,1-dimethylcyclohexane ✓ / Correct. Both methyls share C1.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B08 — NAME → STRUCTURE

Level: Hard | Given name:: 2-bromobutane

1. NAME DECODED

Parent: butane (4 carbons)
Substituents: 2-bromo
Expected formula: C4H9Br

2. BOND-LINE STRUCTURE (THE ALKANE)

4-carbon chain with a -Br group attached at carbon 2.

3. FORMULA CHECK

Counted: C4H9Br -> Matches haloalkane degree of unsaturation (0).

4. BACK-CHECK

Returned name: 2-bromobutane ✓ / Correct.
Numbered to give lowest locant (2) to halogen.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B09 — NAME → STRUCTURE

Level: Hard | Given name:: 1-chloro-3-methylpentane

1. NAME DECODED

Parent: pentane (5 carbons)
Substituents: 1-chloro, 3-methyl
Expected formula: C6H13Cl

2. BOND-LINE STRUCTURE (THE ALKANE)

5-carbon chain with -Cl at terminal carbon C1, and methyl branch at C3.

3. FORMULA CHECK

Counted: C6H13Cl -> Matches.

4. BACK-CHECK

Returned name: 1-chloro-3-methylpentane ✓ / Correct. Numbering starts closest to substituent.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded

CARD B10 — NAME → STRUCTURE

Level: Expert | Given name:: cis-1-chloro-3-methylcyclohexane

1. NAME DECODED

Parent: cyclohexane (6-carbon ring)
Substituents: 1-chloro, 3-methyl (cis)
Expected formula: C7H13Cl

2. BOND-LINE STRUCTURE (THE ALKANE)

Hexagon ring with -Cl at C1 and methyl at C3, both pointing in the same direction.

3. FORMULA CHECK	4. BACK-CHECK
Counted: C7H13Cl -> Matches.	Returned name: cis-1-chloro-3-methylcyclohexane ✓ / Correct. Stereochemistry & locants verified.

VERIFICATION STRIP

[✓] parent length [✓] substituent positions [✓] formula agrees [✓] back-check recorded